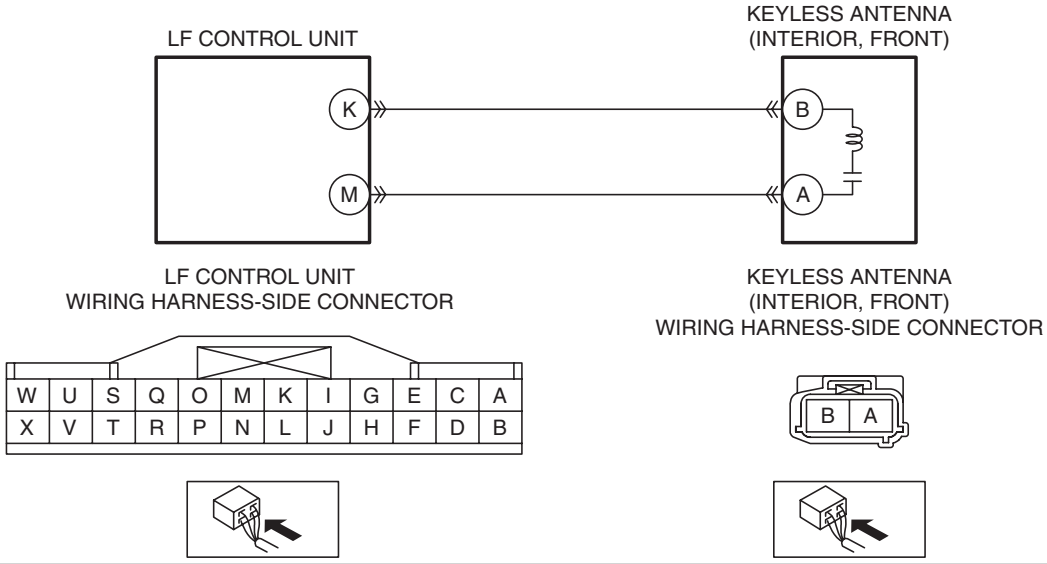
DTC B10C9:1F [ADVANCED KEYLESS ENTRY SYSTEM]

id0902p8026300

Description	Keyless antenna (interior, front) circuit malfunction
Detection condition	• Start stop unit detects an open circuit or short to ground in the keyless antenna (interior, front) circuit with the ignition switched off.
Fail-safe	• Inhibits the door lock/unlock control using the advanced keyless entry system.
Possible cause	• Keyless antenna (interior, front) connector or terminal malfunction • LF control unit connector or terminal malfunction • Short to ground in wiring harness between the following terminals: — LF control unit terminal K—Keyless antenna (interior, front) terminal B — LF control unit terminal M—Keyless antenna (interior, front) terminal A • Open circuit in wiring harness between the following terminals: — LF control unit terminal K—Keyless antenna (interior, front) terminal B — LF control unit terminal M—Keyless antenna (interior, front) terminal A • LF control unit malfunction • Keyless antenna (interior, front) malfunction • Start stop unit malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT KEYLESS ANTENNA (INTERIOR, FRONT) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the keyless antenna (interior, front) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
2	**INSPECT LF CONTROL UNIT CONNECTOR CONDITION** • Disconnect the LF control unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.

Step	Inspection		Action
3	**INSPECT KEYLESS ANTENNA (INTERIOR, FRONT) CIRCUIT FOR SHORT TO GROUND** • Verify that the LF control unit and keyless antenna (interior, front) connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Keyless antenna (interior, front) terminal B — Keyless antenna (interior, front) terminal A • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • LF control unit terminal K—Keyless antenna (interior, front) terminal B • LF control unit terminal M—Keyless antenna (interior, front) terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 7.
		No	Go to the next step.
4	**INSPECT KEYLESS ANTENNA (INTERIOR, FRONT) CIRCUIT FOR OPEN CIRCUIT** • Verify that the LF control unit and keyless antenna (interior, front) connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — LF control unit terminal K—Keyless antenna (interior, front) terminal B — LF control unit terminal M—Keyless antenna (interior, front) terminal A • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • LF control unit terminal K—Keyless antenna (interior, front) terminal B • LF control unit terminal M—Keyless antenna (interior, front) terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.
5	**INSPECT LF CONTROL UNIT** • Inspect the LF control unit. (See LF CONTROL UNIT INSPECTION.) • Is the LF control unit normal?	Yes	Go to the next step.
		No	Replace the LF control unit, then go to Step 7. (See LF CONTROL UNIT REMOVAL/INSTALLATION.)
6	**VERIFY IF MALFUNCTIONING LOCATION IS KEYLESS ANTENNA (INTERIOR, FRONT) DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the advanced keyless entry system using the M-MDS. (See CLEARING DTC [ADVANCED KEYLESS ENTRY SYSTEM].) • Perform the following work: — Switch the ignition off. — Open the front door (driver's side). — Close all the doors and trunk (4SD)/liftgate (5HB). • Retrieve the advanced keyless entry system DTCs using the M-MDS. (See DTC INSPECTION [ADVANCED KEYLESS ENTRY SYSTEM].) • Is the same DTC displayed?	Yes	Replace the keyless antenna (interior, front), then go to the next step. (See KEYLESS ANTENNA REMOVAL/INSTALLATION.)
		No	Go to Step 8.

Step	Inspection		Action
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the advanced keyless entry system using the M-MDS. (See CLEARING DTC [ADVANCED KEYLESS ENTRY SYSTEM].) • Perform the following work: — Switch the ignition off. — Open the front door (driver's side). — Close all the doors and trunk (4SD)/liftgate (5HB). • Retrieve the advanced keyless entry system DTCs using the M-MDS. (See DTC INSPECTION [ADVANCED KEYLESS ENTRY SYSTEM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ADVANCED KEYLESS ENTRY SYSTEM].)
		No	DTC troubleshooting completed.